

DTZZIII-DK200 Series Elevator Load Weighing Controller

Instructions V1.1.2.2.0

1 Technical parameters and types

Type	DK200	DK200-1
Main feature	Over load、full load、light load	
Technology parameter	Power Source	DC24V±10%
	Sensitivity	<0.1 %
	Control return difference	<1 %
	Working temperature	−10 ~ 55℃
	Power consumption of instrument	<10 W
	Overload capacity of sensor	150 %
	Output contactor capacity	0.3A/DC24V; 0.3A/AC220V
Note:	1. The half load of the product is optional, and the relay contact is normally open by default. If there is no special requirement, this function is not available. 2. DK200 is equipped with a single sensor, DK200-1 is equipped with dual sensors	

2 Control instrument appearance size and terminal function

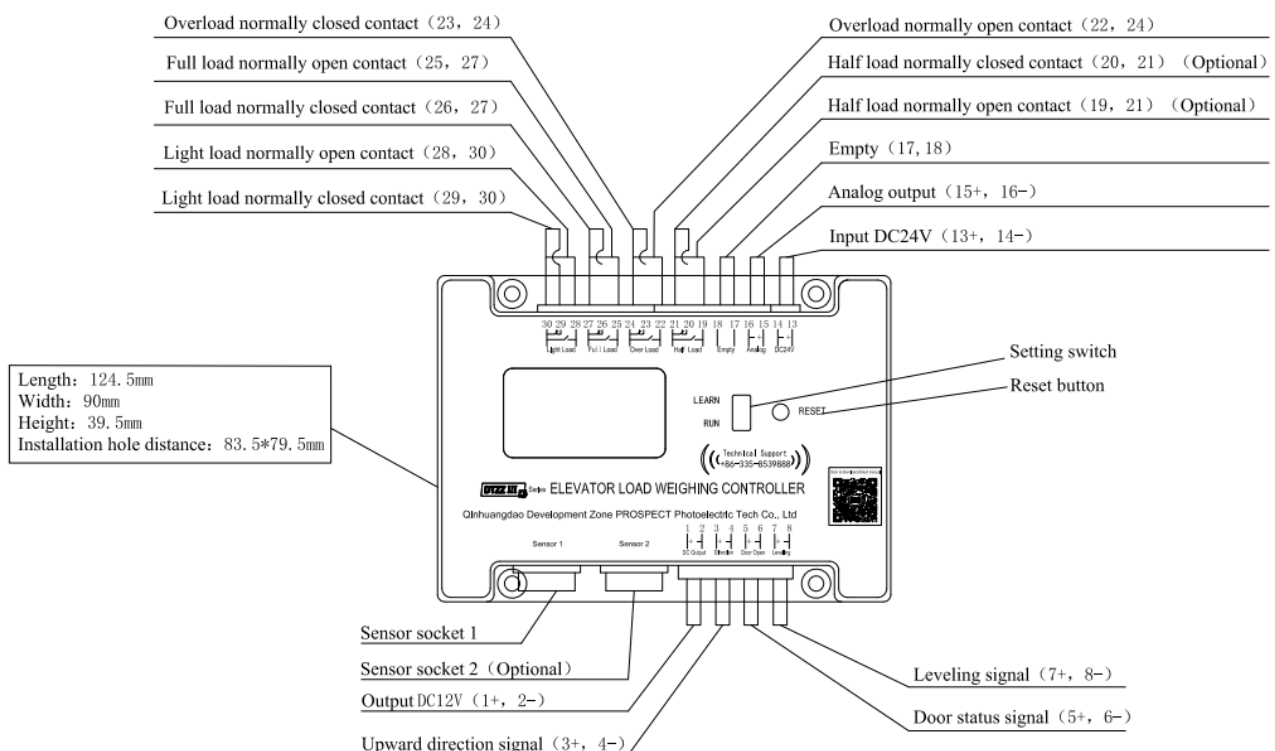


Figure 1 Controller appearance size

Note:

2.1 Each input signal can be a contact signal or a level signal. The wiring method is shown in the figure below. In the contact mode, when the elevator goes up, the contact 1 is closed; when the elevator door

opens, the contact 2 is closed; when the elevator is on a leveling floor, the contact 3 is closed.

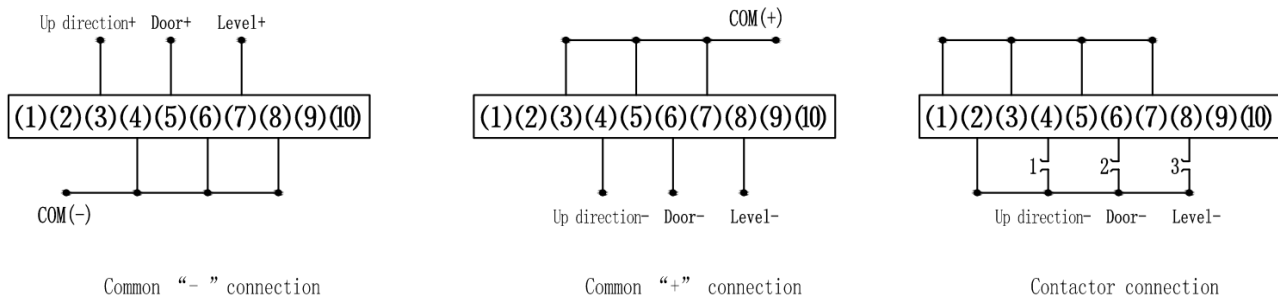


Figure 2 Terminal wiring diagram

2.2 If the input signal of the controller needs to be connected to the double leveling floor, the wiring method is as shown in the figure below.

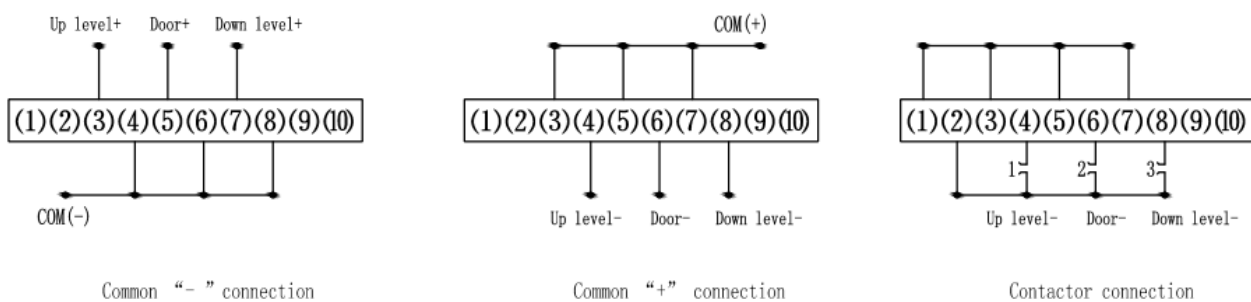


Figure 3 Terminal wiring diagram

3 The installation size and method of controller

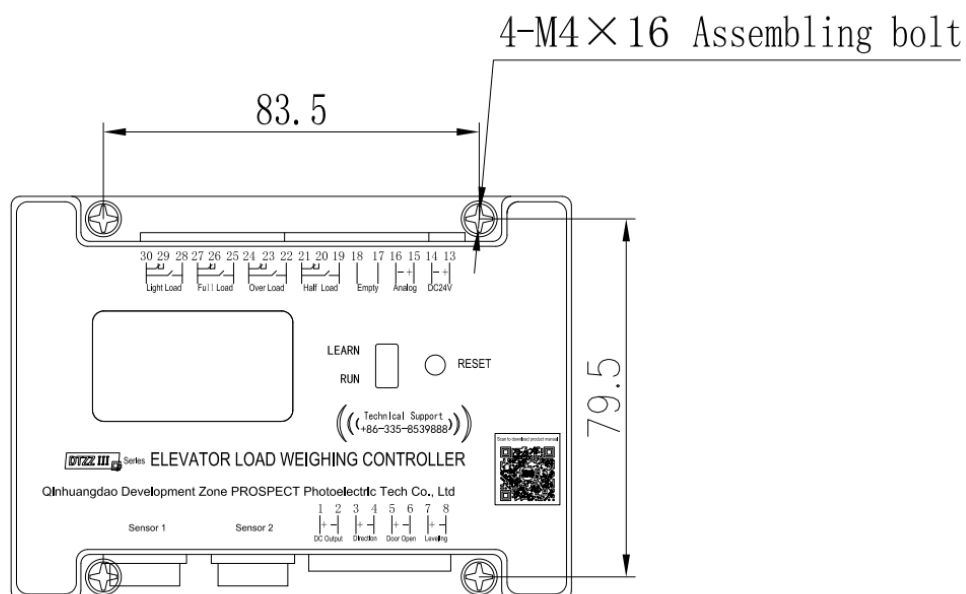


Figure 4 Installation size and method a of controller

3.1 Installation method 1: Use M4 screws to fix the controller to the control cabinet or other places.

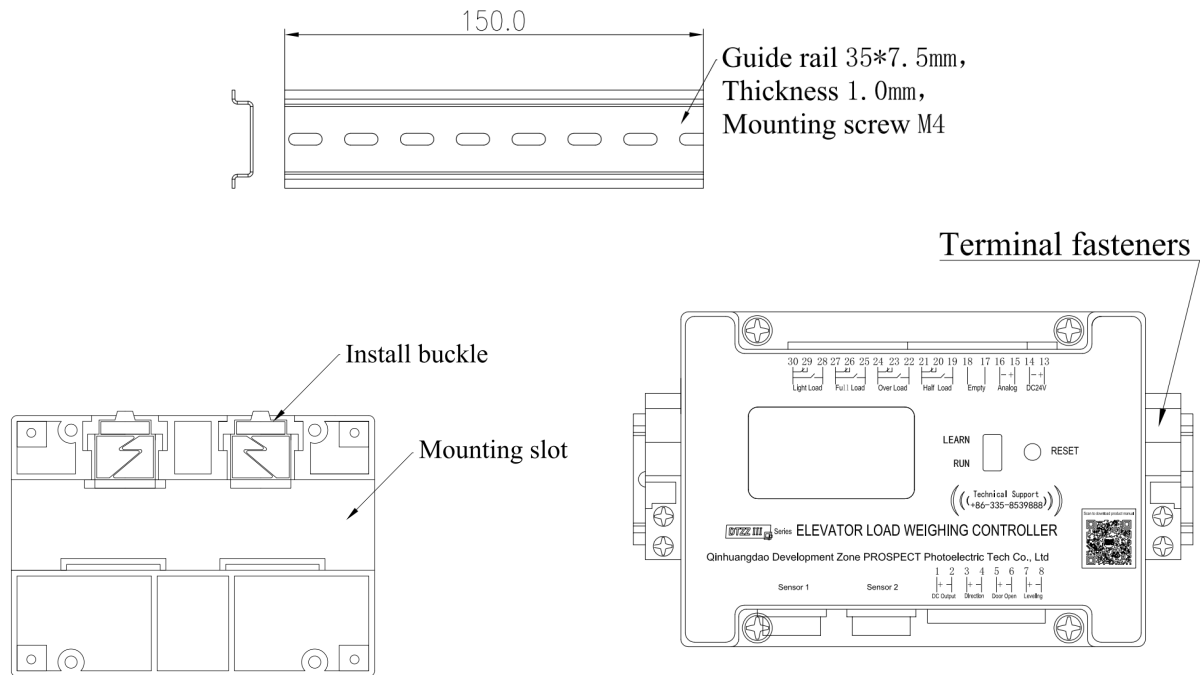


Figure 5 Installation size and method b of controller

3.2 Installation method 2: Use M4 screws to fix the guide rail to the control cabinet or other places. Slide the product installation groove into the guide rail and push the installation buckle to fix it. Then fasten the controller to the guide rail through the terminal fastener.

4 Sensor install picture

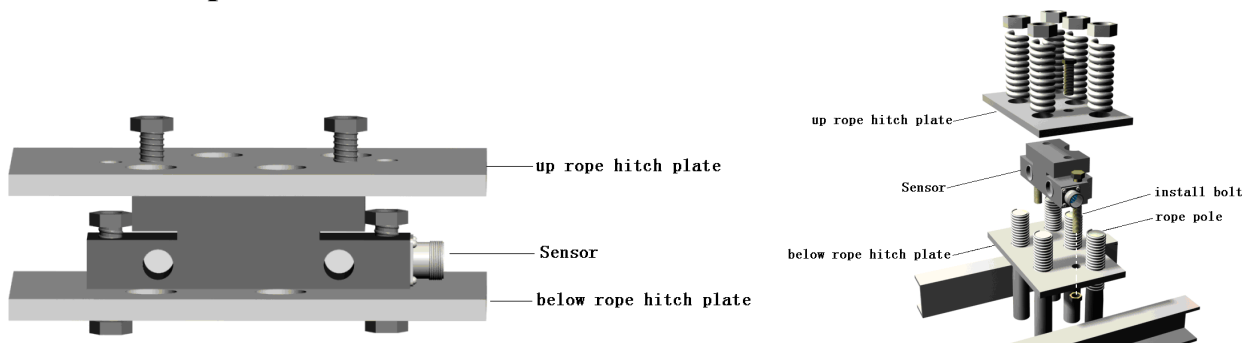


Figure 6 Sensor installation picture

Note:

- 4.1 Additional rope head plate thickness of 20 ~ 25 mm, outside dimensions beyond hole outside around 30 mm.
- 4.2 Sensors placed between two rope hitch plate, connected to the top and bottom rope hitch plate respectively, as shown above.
- 4.3 The part of the connecting bolt between the sensor and the upper rope hitch plate that is screwed into the sensor cannot exceed the lower surface of the sensor connection surface.
- 4.4 Rope rod must avoid contact with the sensor.
- 4.5 Rope rod must avoid contact with the wall of the rope hitch plate hole through which it passes.

5 Initial review

5.1 Power on

Make the “set” switch on running, with power on, the controller will hear one long sound, the display first shows 8888%, light load, half load, full load, overload, then displays --1. When the elevator opens the door, the display shows a percentage value, which increases as the elevator load increases.

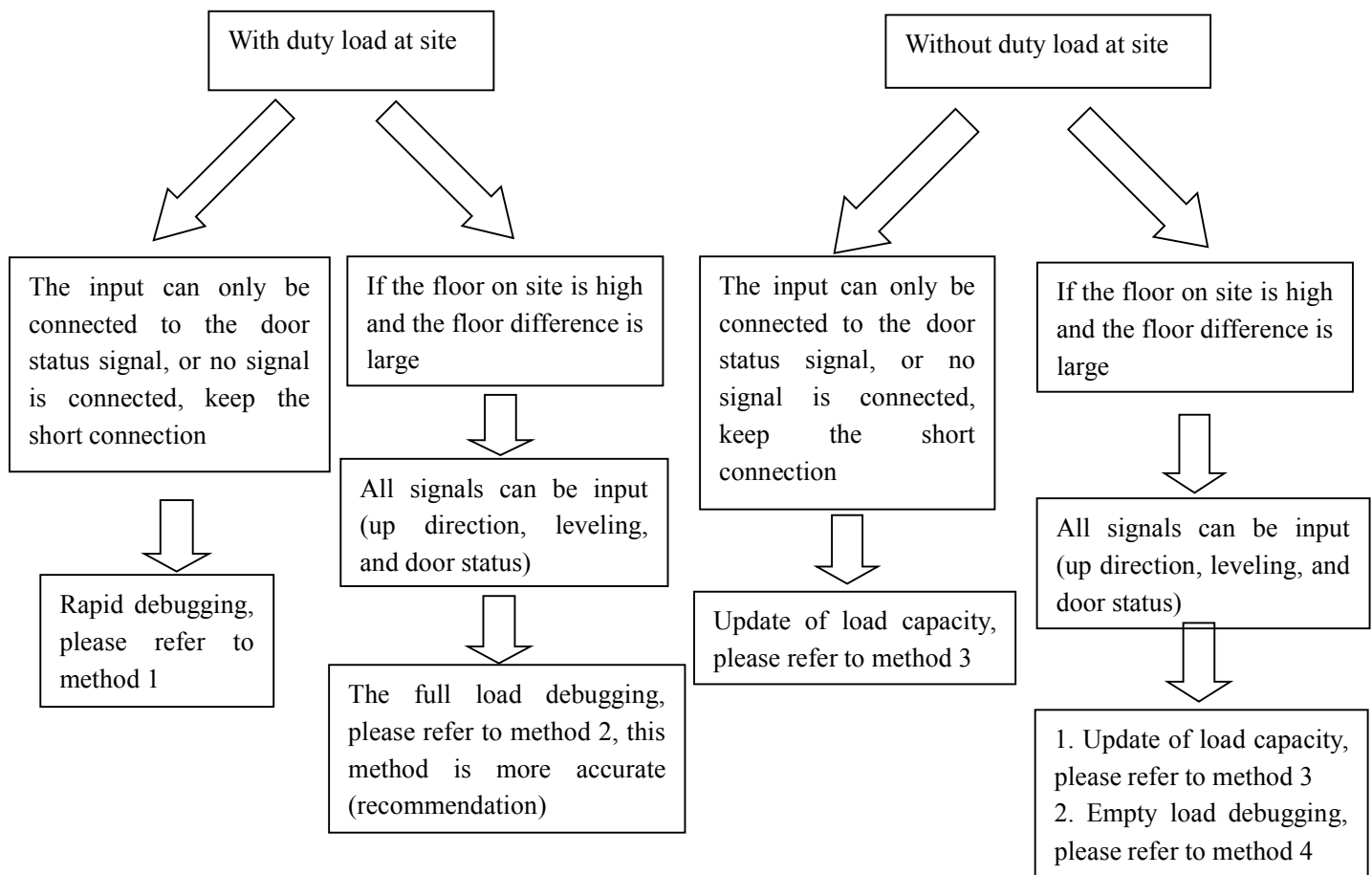
5.2 Reset

5.2.1 Make the “set” switch on running;

5.2.2 Power on and preheating for 30 seconds;

5.2.3 Let elevator empty and stay on the bottom floor, keep the door open. Press the reset button three times in three seconds continuously, at the same time you will hear three short sounds, and then three long sounds, display shows 000%. Now the reset (get rid of the tare) is finished.

6 Debugging (according to the site, choose one method)



6.1 Method 1 Rapid debugging

6.1.1 Connect the door signal, and open the door.

6.1.2 After the initial test, keep the elevator empty and stay on the floor on which the dead weight of the car is light, then reset it.

6.1.3 After reset finished, keep the elevator stay on the floor on which the dead weight of car is heavy, increase the load in the car to 100% of duty load, press the reset switch for more than 5 seconds, when hear a long sound, then the display shows 099% or 100%, release the switch;

6.1.4 Make the controller power off and then power on, the debugging is finished.

6.2 Method 2 Full load debugging

6.2.1 After reset(get rid of the tare), keep the door open, make the load in the car to 100% of duty load, press the reset switch for more than 5 seconds, when hear a long sound, then the display shows 099% or 100%, release the switch;

6.2.2 Press the set switch up to learning position, you will heard 3 long sounds and 1 short sound. Close the door, make the elevator stay on every floor and open the door from the bottom floor to the top floor one by one.(Every time the elevator stops at a floor, the controller sends a self-learning response)

6.2.3 When reach the top floor, press the reset switch 5 times in 5 seconds continuously, at the same time you will hear 5 light sounds, and then 5 long sounds. Now make the elevator stay on every floor and open the door from top floor to bottom floor one by one(Every time the elevator stops at a floor, the controller sends a self-learning response).

6.2.4 After reach the bottom floor, keep the door open, press the set switch down to running position. The instrument goes into normal operation, and the display shows 100%.

6.2.5 Make the controller power off and then power on, the debugging is finished.

6.3 Method 3 Update of load capacity

6.3.1 The elevator stops at the ground floor and opens the door, clear without load (get rid of the tare)

6.3.2 Load $\geq 20\%$ of the load.

6.3.3 Power off the controller, press the reset button, and turn on the power again, turn the setting switch up to the position of <Learning>. After hearing three long beeps, press the reset button once, accompanied by a light beep.

6.3.4 Calculate the theoretical set value of the load according to the recorded loading load. Theoretical set value = loading load $\div$ rated load $\times 100\%$.

6.3.5 Press the reset button to adjust the displayed original set value to the calculated theoretical set value. (The set value cycles between 000-100).

6.3.6 Turn the setting switch back to the <Run> position, and then power on the controller again after power off, and the debugging is completed.

6.3.7 Example: The rated load of the elevator is 1000kg. When the elevator adds a load of 200kg, the theoretical set value of the device should be $200 \div 1000 \times 100\% = 20$. If the displayed value is 23%, through the reset button, manually adjust the display of the digital tube to 20.

6.4 Method 4 Empty load debugging

6.4.1 First please reset (get rid of the tare) the instrument.

6.4.2 Press the set switch up to learning position, you will heard 3 long sounds and 1 short sound. Close the door, make the elevator stay on every floor and open the door from the bottom floor to the top floor one by one.(Every time the elevator stops at a floor, the controller sends a self-learning response)

6.4.3 When reach the top floor, press the reset switch 5 times in 5 seconds continuously, at the same time you will hear 5 light sounds, and then 5 long sounds. Now make the elevator stay on every floor and open the door from top floor to bottom floor one by one(Every time the elevator stops at a floor, the controller sends a self-learning response).

6.4.4 After reach the bottom floor, keep the door open, press the set switch down to running position. The instrument goes into normal operation, and the display shows 000%.

6.4.5 Power on the controller again after power off, the debugging is finished.

7 Set the light load point

When leaving factory, the default setting of light point is 10%. If you need reset the light point, please operate as follow:

When the instrument is in normal working state (elevator is no-load), press the reset button, and after hearing a short beep within 2S, turn the setting switch up immediately, release the reset button, and hear three long beeps, enter the adjustment state, press the reset button to adjust the light load point. The displayed value will cycle between 002-050. After adjusting to the desired light load point, pull down the setting switch to exit the state of setting the light load point.

8 Fault handling

In normal operation, if the instrument displays 666, overload relay output alarm status, please check whether the connection between sensor and controller is well.

Note: when the elevator opens the door, the instrument test the duty load, and output the contactor signal and electricity signal. When the elevator closes the door, the electricity signal will not change.

If any question, or any faulty with controller, please contact us in time.

When the product is scrapped, the relevant disposal regulations related to the list of hazardous wastes must be strictly implemented.

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